Duggireddy Hitesh Pranav Reddy

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EDUCATION

B.Tech, Electronics and Communication Engineering — Sept 2023 – July 2027

PES University, Bengaluru — CGPA 8.37 / 10 (as of 6th semester)

High School — 2011 – 2023

Sri Chaitanya College of Education, Bengaluru — Class XII 91.4% · Class X 83.2%

Relevant coursework Computer-Aided Digital Design (digital logic, HDL, VHDL, Verilog, SystemVerilog, FPGA); VLSI; Analog Circuit Design (incl. MOSFETs); Computer Organization & Design and High-Performance Computing (RISC-V processors, GPUs); Architectures for Hardware Acceleration (elective); Signals & Systems and Digital Signal Processing; Digital Communication; Machine Learning · Artificial Neural Networks · Deep Learning (electives)

EXPERIENCE

Hardware Student Trainee — Samsung Semiconductor India Research — Jan 2027 – Jun 2027

Research Intern — C-ISFCR, PES University — Jun 2025 – Aug 2025

- Developed an ML-based Intrusion Detection System for automotive CAN bus networks in Python (built and run on Linux — Ubuntu and Kali).
- Built a real-time Python firewall for CAN traffic filtering and anomaly detection.
- Trained and evaluated ensemble ML models with scikit-learn; analyzed misclassifications to identify and address root causes of false detections.
- Built a Tkinter test harness for CAN message injection, traffic monitoring, and analytics; used it to exercise the system against simulated attack traffic and validate detection behaviour.
- Collaborated with researchers to validate system performance across multiple attack scenarios and presented findings to the supervising team.

Student Head (previously Teaching Assistant, Core Member) — Joy of Engineering Lab (JoEL), PES University

Oct 2024 – Present

- Mentor 6+ student teams on IoT and embedded-systems projects, translating open-ended problems into structured, scoped plans.
- Lead technical/backend planning and logistics for HackeZee 2025 and Roadshow 2026, coordinating volunteers, mentors, vendors, and technical teams across multiple sub-teams under deadlines.
- Designed, built, and maintain the official JoEL website (Next.js, TypeScript, Tailwind CSS) with an interactive inventory management system for equipment tracking and availability monitoring.
- Ran cross-browser / responsive testing and fixed reported issues from student and faculty users.
- Collaborate with faculty and student organizations to execute technical initiatives.

Digital Marketing & Web Designer (Freelance) — Shree Sports and Ortho Clinic — Ongoing

- Designed, built, and maintain the clinic's official website.
- Run digital marketing for the clinic (content, online presence).

RTL DESIGN & VERIFICATION PROJECTS

AES-128 Encryption Core — SystemVerilog · Vivado · FPGA

- Designed and verified a synthesizable AES-128 encryption core (iterative, FSM-driven datapath) in SystemVerilog, targeting hardware/RTL interview prep for networking-hardware control-path design.
- Verified bit-exact output in behavioral simulation against the official FIPS-197 test vector.
- Synthesized out-of-context for a Xilinx Artix-7, analyzing LUT/register utilization and closing static timing at 100MHz — root-caused a routing-delay bottleneck from a high-fanout FSM register and applied a fanout-replication fix, improving worst negative slack from 1.068ns to 3.704ns.

Single-Cycle RISC-V (RV32I) Processor — R-type Instructions — SystemVerilog · Vivado · RV32I

- Implemented a single-cycle RV32I datapath (PC, instruction memory, register file, ALU, control unit) in SystemVerilog, decoding all 10 R-type ops via real opcode/funct3/funct7 encoding — including disambiguating ADD/SUB and SRL/SRA, which share funct3 and differ only in funct7.
- Designed a 10-operation ALU with correct signed vs. unsigned semantics (SLT vs. SLTU, SRA's sign-extension vs. SRL) and a register file with x0 hardwired to zero, combinational reads, and synchronous writes.
- Verified in Vivado behavioral simulation via a backdoor-preloaded testbench (register values can't be loaded through this instruction subset alone): 10/10 destination registers matched expected values, PASS=10, FAIL=0.

Layered SystemVerilog Testbench for a 4-bit Multiplier

SystemVerilog · Functional Coverage · Vivado

- Built a reusable, layered verification environment (generator, driver, monitor, scoreboard) as independent SystemVerilog classes communicating via mailboxes, run concurrently via fork/join_none, around a 4-bit x 4-bit combinational multiplier DUT.
- Combined dist-weighted constrained-random stimulus with directed corner-case tests to guarantee every functional-coverage cross-bin is hit regardless of random luck.
- Self-checked against an independent reference model in the scoreboard; achieved 100% functional coverage.

Hardware Trojan Detection & ATPG Analysis

Python · Verilog · ATPG · SCOAP · DFT

- Engineered a Python automation framework to parse gate-level Verilog netlists and programmatically insert Hardware Trojans / faults into ISCAS-85 benchmark circuits.
- Generated fault lists and test vectors and ran Automatic Test Pattern Generation (ATPG) with Atalanta, measuring fault coverage to quantify how effectively generated tests detect defects.
- Conducted structural graph and toggle-rate analysis using SCOAP testability metrics to compare clean vs. Trojan-inserted circuits and identify hard-to-detect cases.
- Visualized suspicious gate-level activity to evaluate test-vector effectiveness.

CMOS Inverter & D Flip-Flop — Schematic & Layout

Cadence Virtuoso

- Designed transistor-level schematics for a CMOS inverter and a D flip-flop, sizing NMOS/PMOS devices and verifying via analog simulation.
- Created full-custom physical layouts and closed the loop with DRC and LVS — clean, rule-compliant, schematic-matched layouts.

OTHER PROJECTS

End-to-End Autonomous Driving: Architecture Ablation Study

Python · PyTorch · scikit-learn · CNN

Designed a controlled ablation study in the CARLA simulator to isolate the contribution of spatial vs. temporal architectures in predicting throttle, steer, and brake from camera input, holding data and training constant.

Automotive ML Intrusion Detection System

Python · C · scikit-learn · Tkinter

Designed a real-time intrusion detection system for CAN bus traffic as modular components for message parsing, feature extraction, and classification.

Audio Signal Denoising & Spectral Analysis

Python · MATLAB · NumPy · SciPy

Built a Python/MATLAB signal-processing pipeline to analyze large sampled-signal (noisy audio) datasets in both time and frequency domains.

Motorsport Telemetry & Sensor Fusion Dashboard

Python · IMU · GPS · Sensor Fusion

Built a real-time race-telemetry dashboard streaming live IMU acceleration (Ax, Ay, Az), GPS position, and a 720p MJPEG video feed from the vehicle.

PESU Research Match

Flask · PyTorch · Sentence-Transformers · Docker

Built a hybrid semantic search engine combining dense vector retrieval (BGE-Large) and BM25 lexical search for faculty recommendation / matching.

SKILLS

Digital design & verification	RTL design, functional verification, constrained random stimulus, functional coverage, ATPG, DFT, FSM design, timing analysis, logic synthesis, UART/serial communication, schematic & layout, CMOS devices
Languages	Verilog, SystemVerilog, Assembly (RISC-V), C, C++, Python, MATLAB
Tools	Xilinx Vivado, Cadence Virtuoso, Atalanta, Git & GitHub, Linux, Jupyter, Docker
ML & signal processing	PyTorch, scikit-learn, NumPy, SciPy, pandas, CNNs, GRUs, ensemble learning, ablation studies, FFT/DFT, FIR/IIR filtering, sensor fusion
Web	TypeScript, JavaScript, React, Next.js, Tailwind CSS

ACHIEVEMENTS & CERTIFICATIONS

- 5th overall and 2nd-lightest vehicle, Ignition 1.0 — 18-hour telemetry hackathon (Team Vegavath PESU, sponsored by Ather Energy)
- Technical lead for HackeZee 2025 and Roadshow 2026 (PES University engineering events)
- Python for Beginners: Learn Python Programming (Python 3) — Udemy, Jason Cannon, Apr 2024